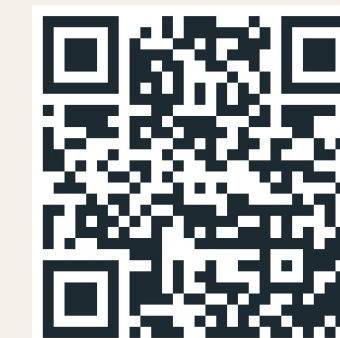


Learning Normal Representations for Blood Biomarkers



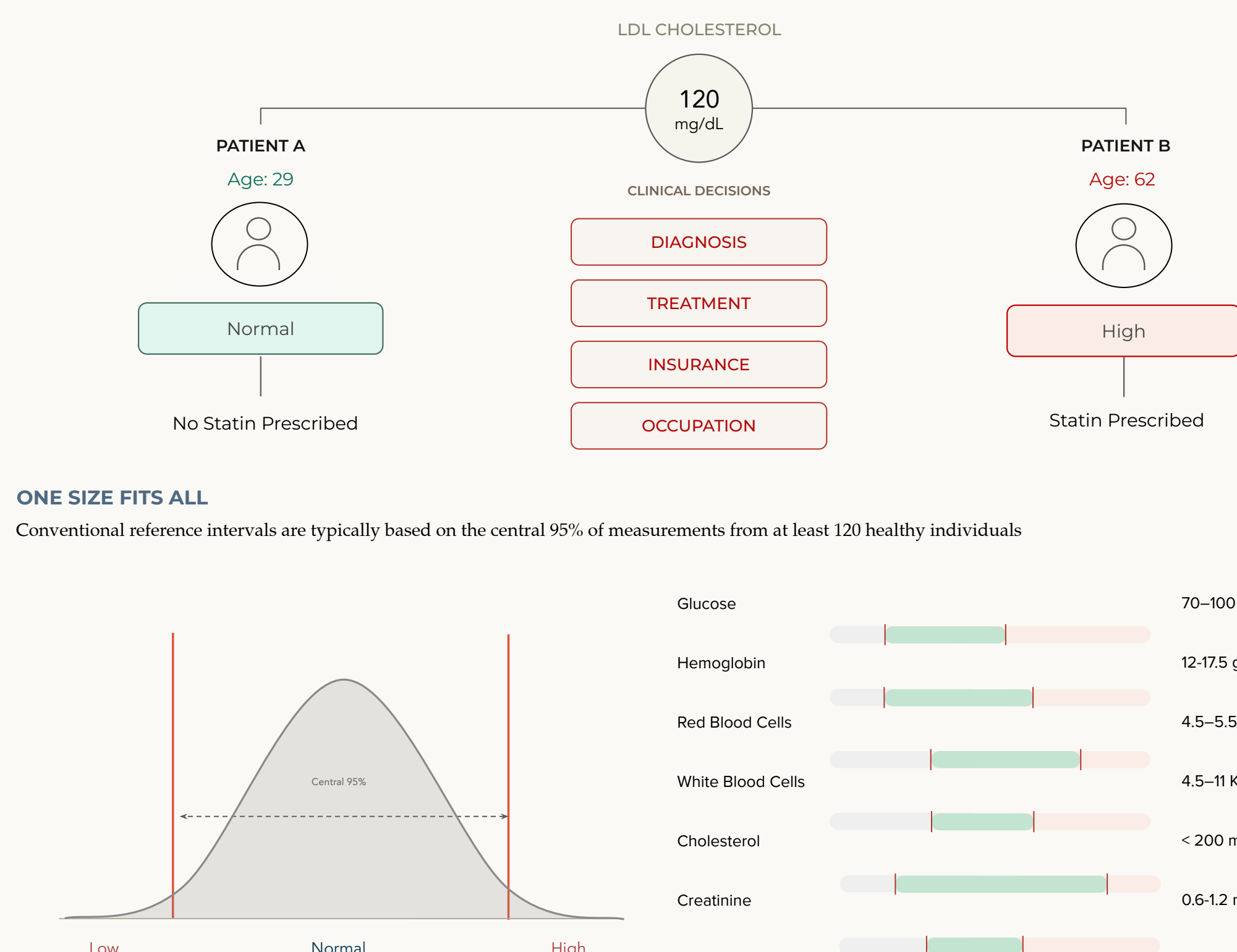
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Harvard Medical School; ² Brigham and Women's Hospital; ³ Johns Hopkins University; ⁴ Boston Children's Hospital; ⁵ Harvard Medical School and Clalit Research Institute; ⁶ Clalit Research Institute; ⁷ Ben-Gurion University; ⁸ Co-senior authors

BACKGROUND AND MOTIVATION

Population reference ranges can mask meaningful personal change

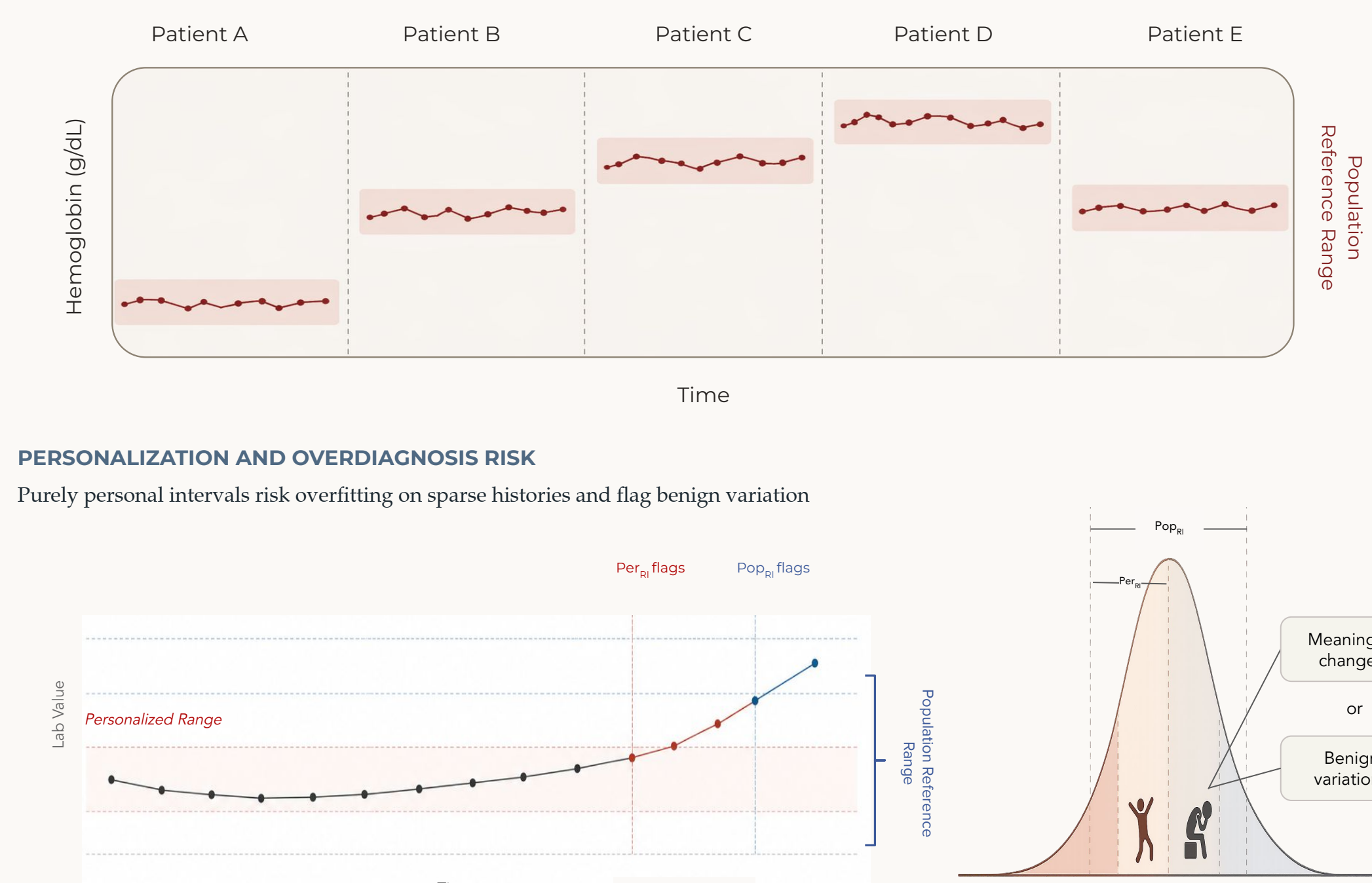
CONTINUOUS TO CATEGORICAL

The same value can result in different clinical decisions based on interpretation



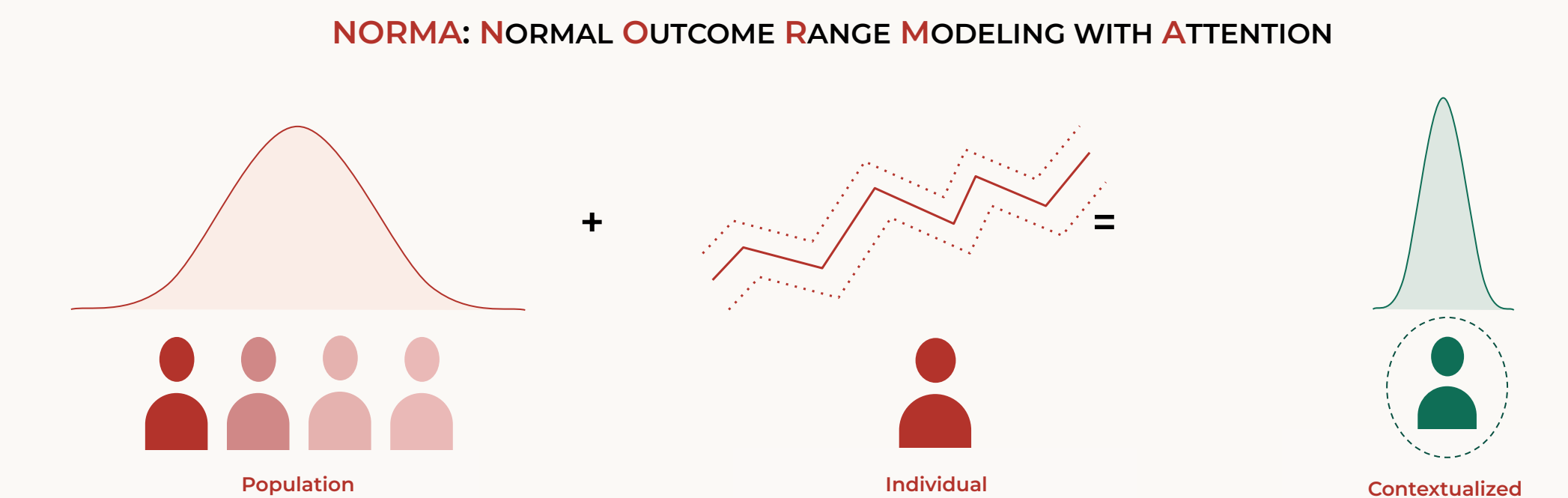
STABLE PATIENT TRAJECTORY

Blood indices often fluctuate within a narrow interval in a consistent and patient-specific manner



CONTEXTUALIZED REFERENCE INTERVALS

NORMA anchors a personal trajectory to population expectations

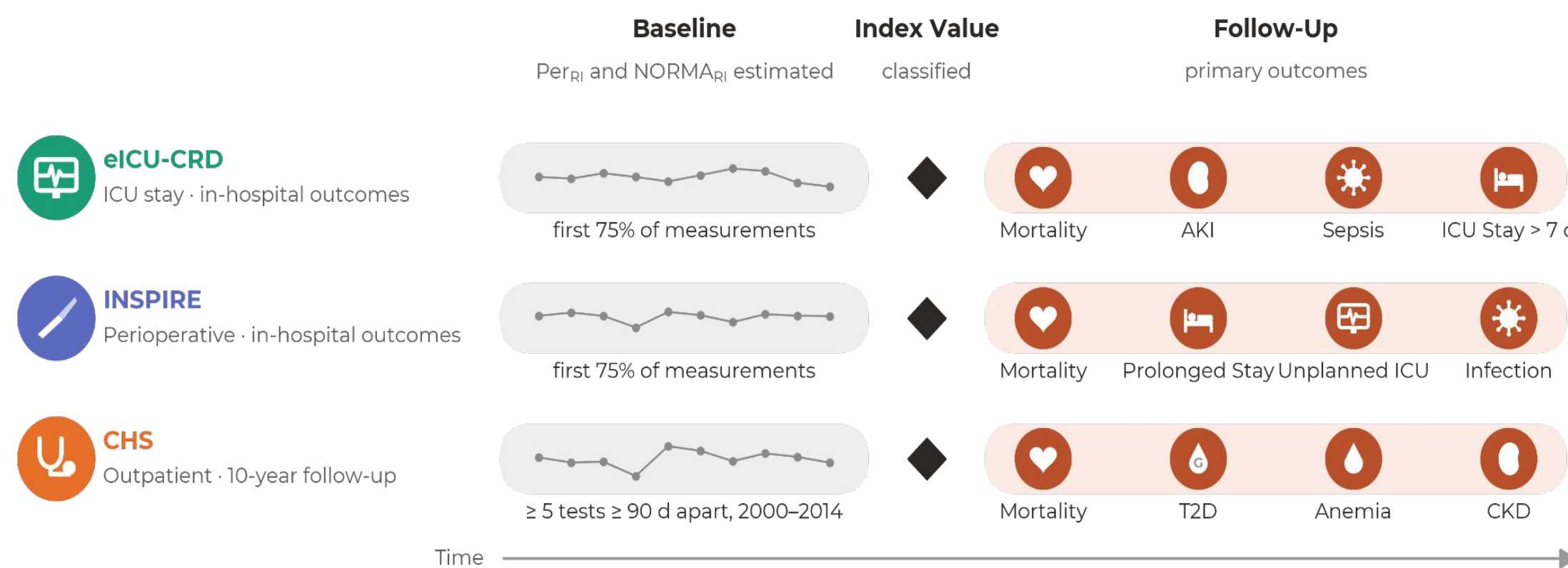


AIMS

Can population context stabilize personalized intervals while preserving clinically meaningful signal?

STUDY TIMELINE AND OUTCOMES

Evaluation of reference intervals in three care settings



1 Routine blood biomarkers are stable and individualized

COHORT OVERVIEW

Five longitudinal cohorts, three countries, 1.8M patients

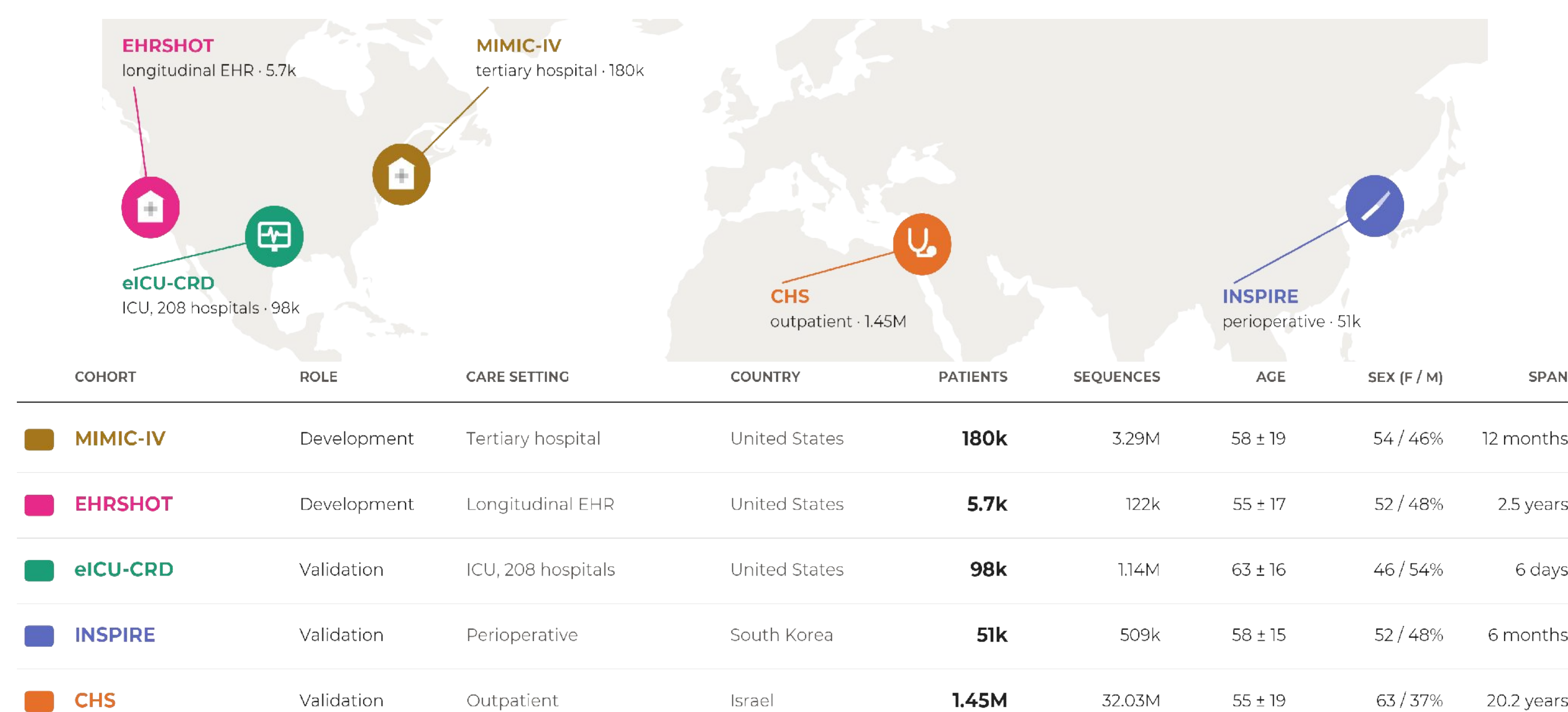


Fig 1 | Study populations. NORMA was developed in MIMIC-IV and EHRSHOT, and externally validated in CHS (outpatient, 1.45M), eICU-CRD (ICU, 208 hospitals, 98k), and INSPIRE (perioperative, 51k).

2 NORMA calibrates to individual trajectories

NORMA ARCHITECTURE

A sequence decoder predicts the distribution of the patient's next lab value given their own history

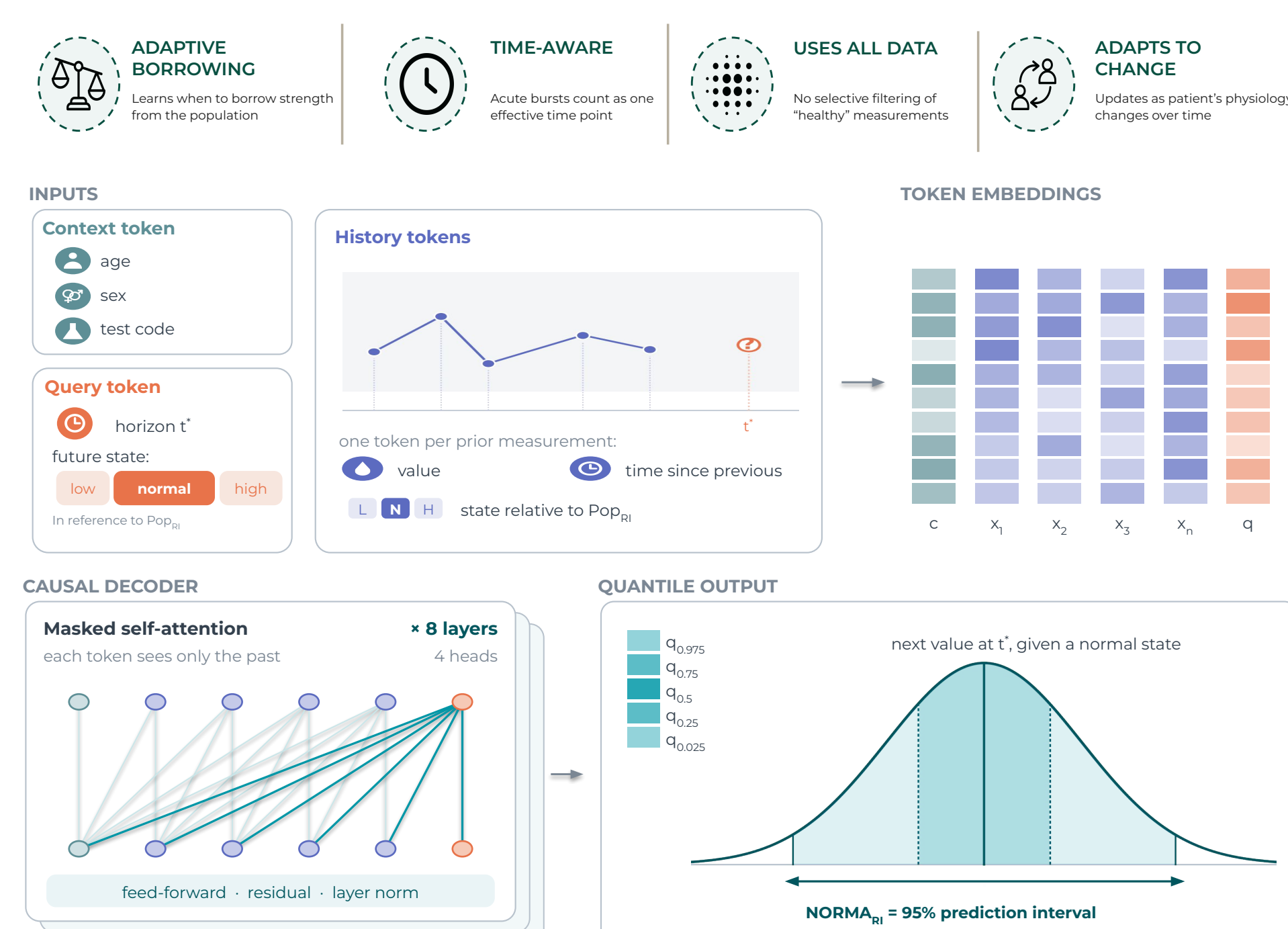


Fig 4 | NORMA architecture. A context token, one token per prior measurement, and a query token specifying a future normal state and horizon are passed through a causally masked transformer decoder. Quantiles are fit with a pinball loss at levels 0.025, 0.25, 0.5, 0.75, and 0.975, and NORMA₉₅ is defined as the 95% prediction interval. The model was trained on 3.4M sequences from MIMIC-IV and EHRSHOT.

3 NORMA detects clinically meaningful changes in patient baselines

FUTURE ABNORMAL VALUES

NORMA's flags best anticipate abnormalities defined by population reference ranges

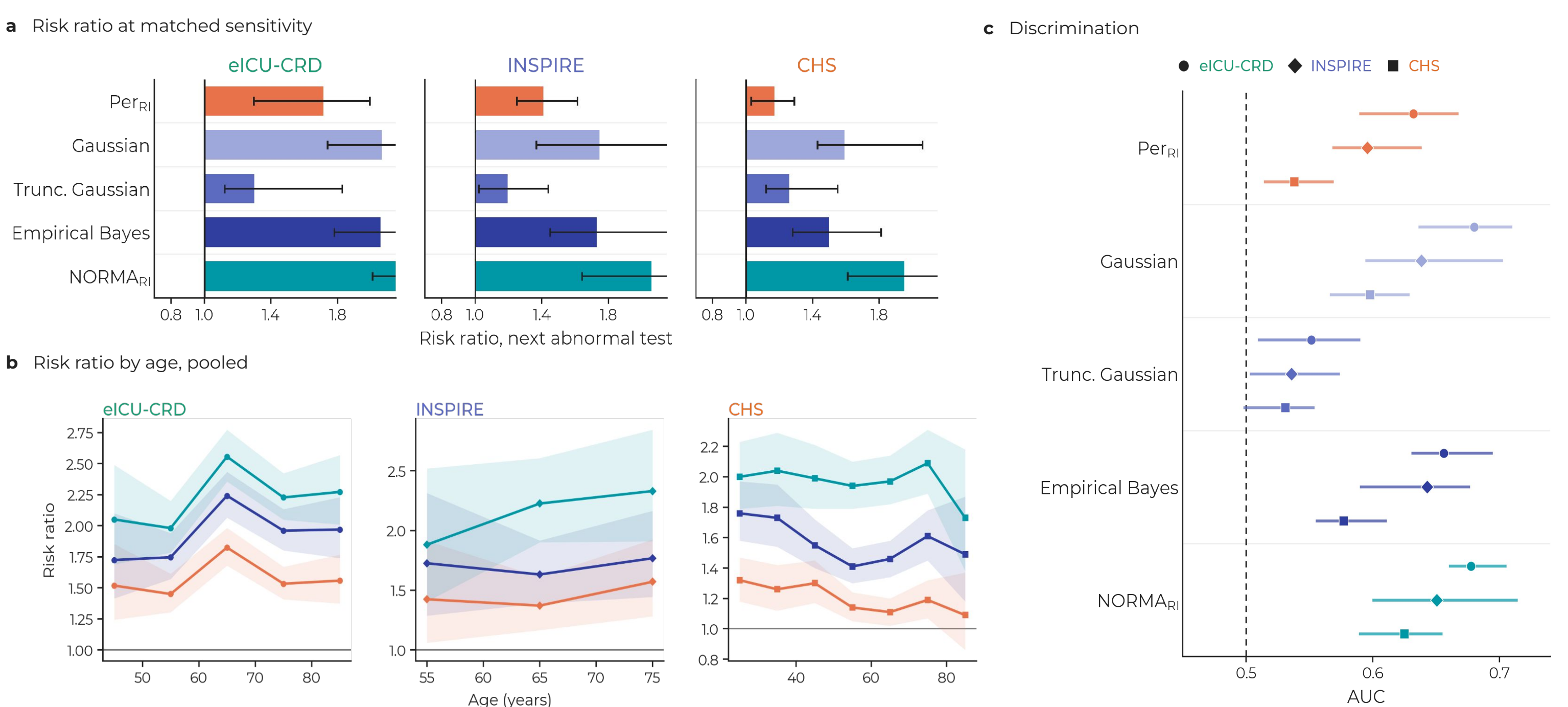


Fig 7 | Anticipating abnormal values. a, Risk ratio that a flagged test inside Pop_R is followed by a Pop_R-abnormal test within 7 days, at matched sensitivity (0.2). Values are the median and interquartile range across analytes. b, The same risk ratio by age band, pooled across analytes, one panel per cohort; CHS values are transcribed from pooled tables and carry no confidence interval. c, Area under the receiver operating characteristic curve for the same endpoint, by method and cohort.

INDIVIDUALITY

Within-person variation accounts for ~40-70% of between-person variation

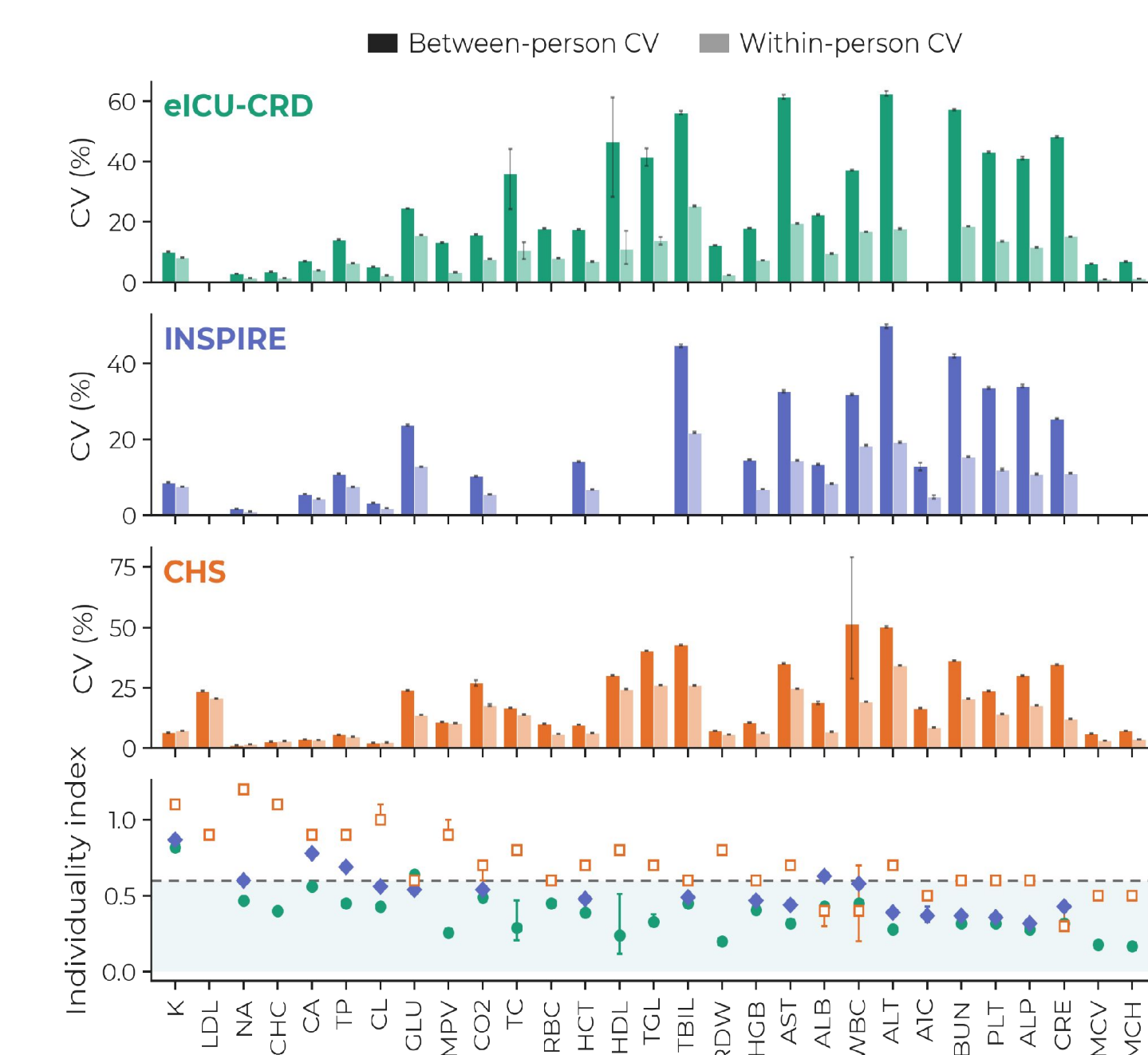


Fig 2 | Within-person and between-person variation. Top three rows, between-person and within-person coefficient of variation per analyte, one row per validation cohort. Analytes are ordered by mean index; hollow markers denote CHS values, rounded to one decimal. Error bars denote 95% confidence intervals.

MORTALITY BY VALUE

Monotonic and U-shaped association between lab values and all-cause mortality

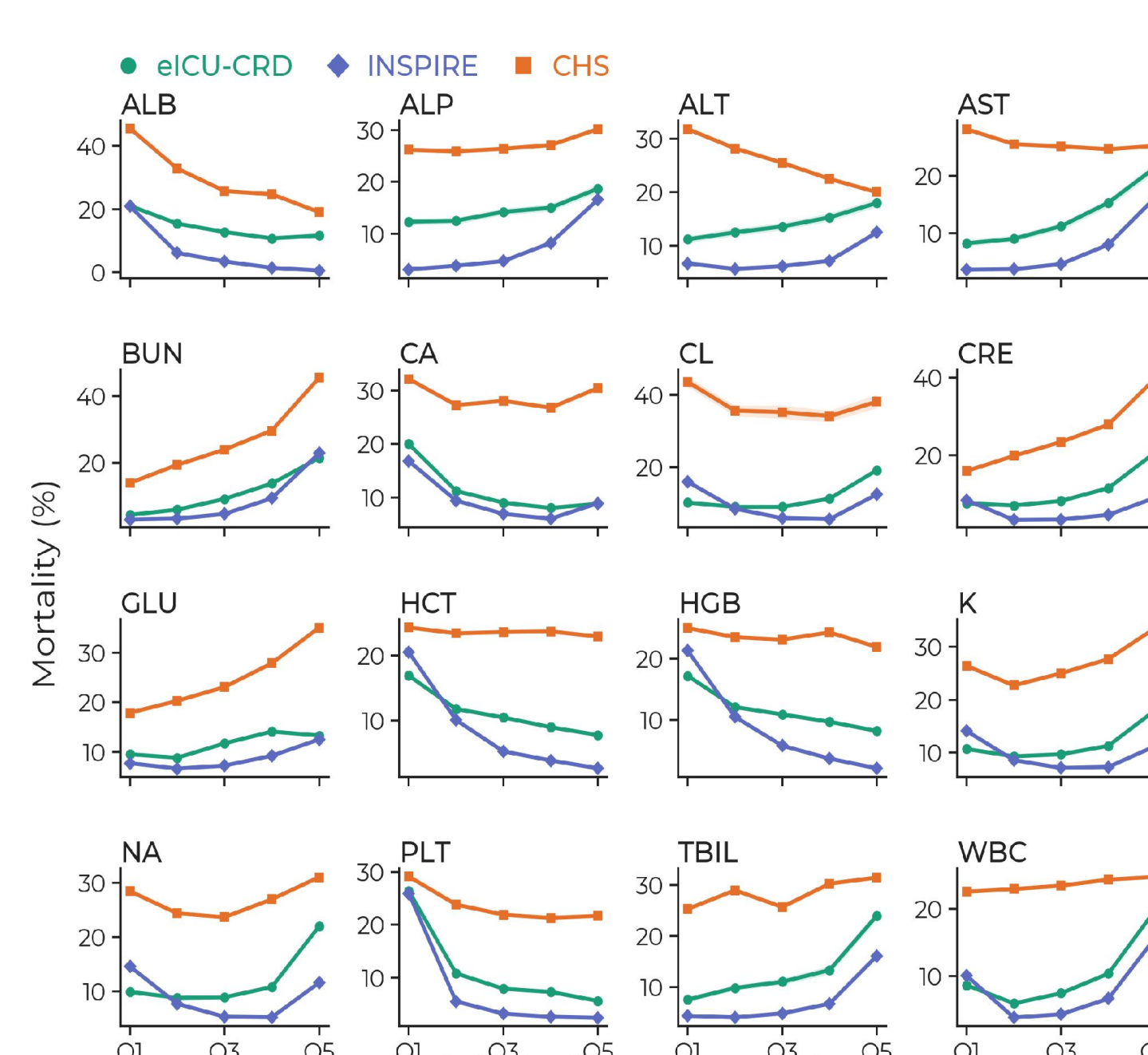


Fig 3 | Mortality by measured value. Observed all-cause mortality rate by quintile of the measured value, for the sixteen analytes reported by all three validation cohorts. Associations are monotonic for some analytes and U-shaped for others. Error bars denote 95% confidence intervals.

FORECASTING AND CALIBRATION

NORMA predicts the next value best and holds close to 95% coverage

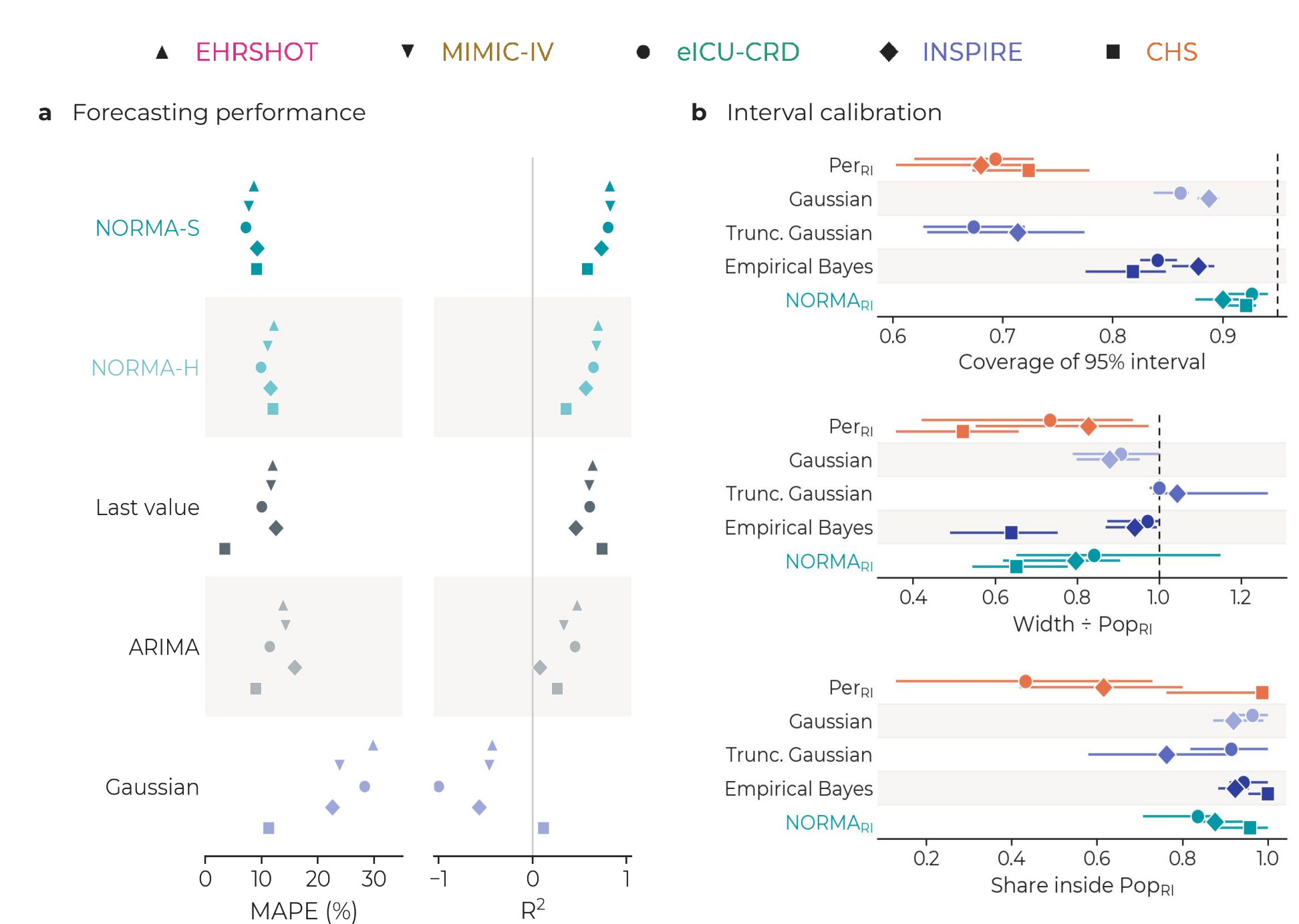


Fig 5 | Next-value accuracy and interval calibration. a, Next-value forecasting accuracy, averaged across analytes. NORMA-S is given the state of the next value (low, normal, or high); NORMA-H uses the measurement history alone and is directly comparable to the baselines. b, Coverage of the 95% interval, interval width relative to Pop_R, and share of measurements inside Pop_R. Values are the median and interquartile range across analytes.

INTERVAL WIDTH AND FLAG RATES

Intervals borrow strength from the population when individual history is noisy

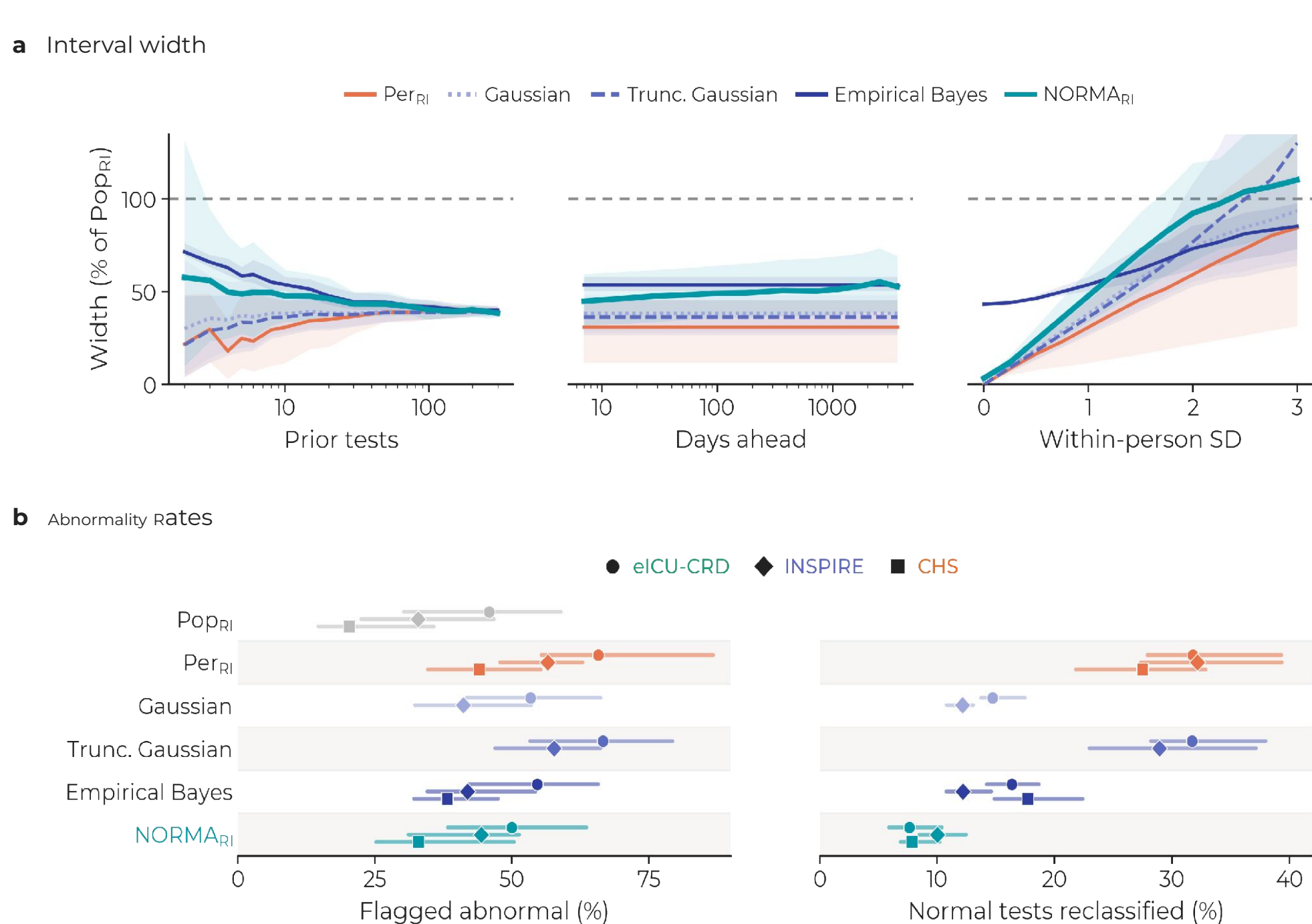


Fig 6 | Interval width and abnormality rates. a, Sensitivity of median prediction interval width on synthetic histories to history length, forecast horizon, and within-person variability. b, Share of tests flagged abnormal, and share of Pop_R-normal tests reclassified as abnormal. Values are the median and interquartile range across analytes.

CLINICAL OUTCOME RISK

NORMA flags laboratory values that are strongly associated with clinical outcomes and produces far fewer false alarms

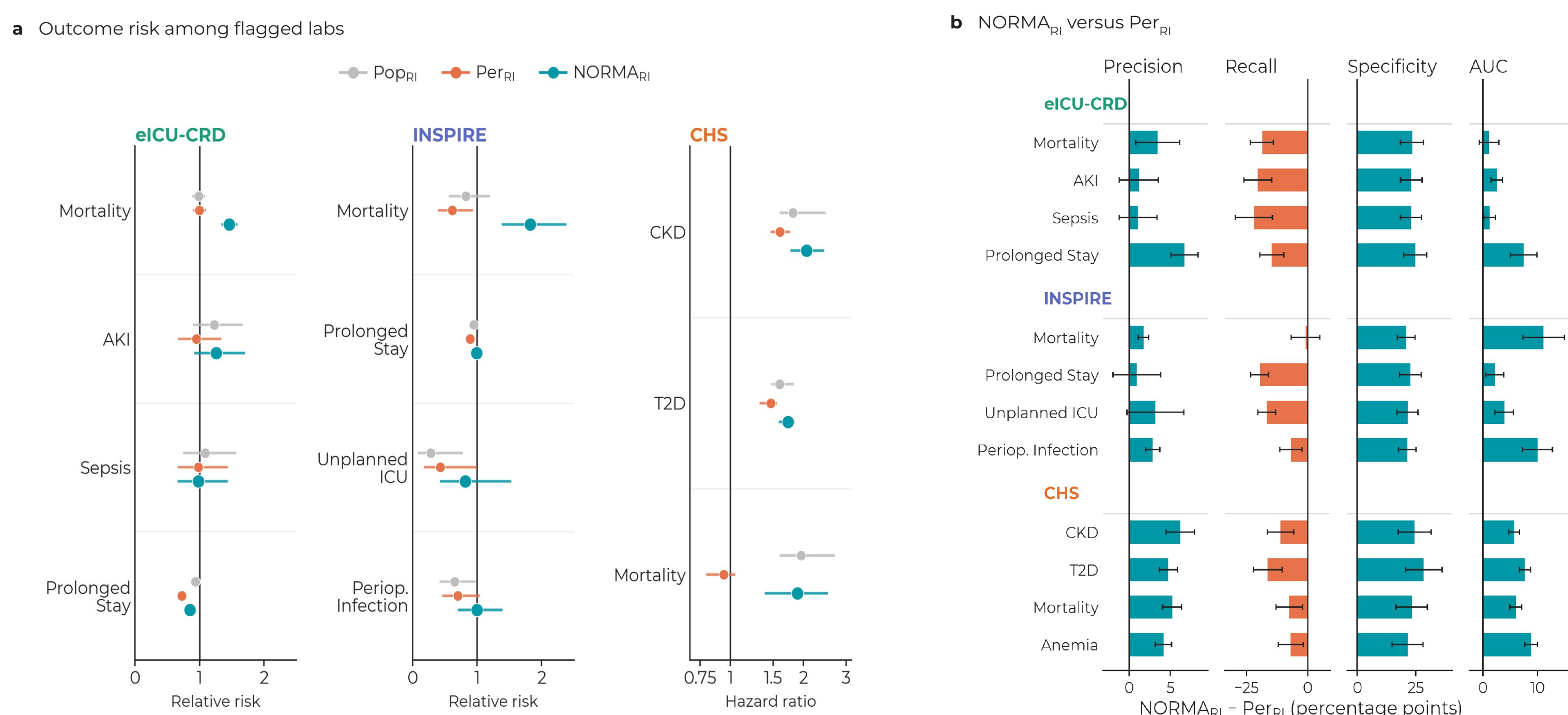


Fig 8 | Clinical outcome risk among flagged tests. a, Relative risk of each outcome in eICU-CRD and INSPIRE among the 10% of patients flagged most strongly by each method in the first 24 h, restricted to tests inside Pop_R. CHS, which is followed for ten years rather than a single stay, is shown as the Cox hazard ratio associated with an abnormal flag. b, Difference between NORMA₉₅ and Per_R in percentage points, for precision, recall, and specificity at each method's own threshold, and for the AUC of the deviation score. Values are the mean across analytes. Error bars denote 95% confidence intervals.